



Certified TestStand Developer Examination

Examinee: _____ Date: _____

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- A computer with a standard installation of LabVIEW, TestStand, and the software libraries required for the exam is the only reference allowed for the examination. Externally developed code or third party tools are not allowed in the exam. You may use any design patterns, templates, and examples available in the development environment as a guide/resource for the application development.
- The application must be specifically developed for the exam submission.
- The required software packages for the application are provided to you installed on the local machine under the users\public\documents directory. You **must** maintain the folder hierarchy and use these components to develop your application. Solutions that do not use the hierarchy or the given components are not graded.
- Do **not** rename any of the Vis, libraries or any other provided material.

IMPORTANT:

- **When you complete the exam, <Instructions on exam completion here>**

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Section I: General Requirements

The Certified TestStand Developer exam tests your ability to develop a TestStand Sequence.

The application must do the following:

- Function as specified in Section II of this document
- Conform to TestStand style and documentation standards ([Guide to Developing Effective, Maintainable TestStand Sequences](#))
- Be created expressly for this exam using step types and functions available in TestStand, along with code modules provided. Templates, examples, or code developed outside the bounds of this exam are not acceptable for use in the application
- Close all opened references and handles where used
- Be easily scalable to add more states/features without manually updating the hierarchy
- Pass Sequence Analyzer without errors.

Grading:

The CTD exam consists of a total of 100 points which are allocated as below. Exam passing grade is 70%

- Functionality **(100 points)**

Time allocated for the exam: 4 hours

Objective

Design an automated sequence using NI TestStand to test a SP25 Circuit Card as per the specifications detailed below. You must use the provided APIs and code to develop your application.

SP25 Overview

The SP25 is a Maximum Power Point Tracking (MPPT) Charge Controller. It is designed for use in solar panel systems where the solar panels feed power to the SP25, which then adapts the power in order to charge batteries. The diagram below shows the SP25 as it would be used in a complete system.

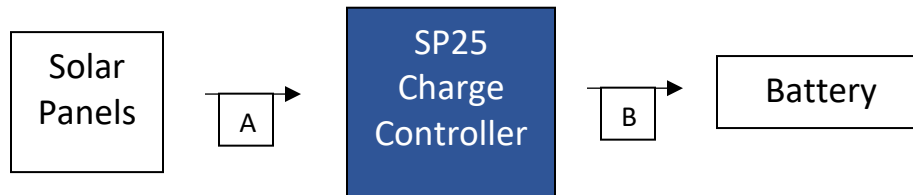


Figure 1- SP25 shown in complete system view.

- A. Power input from the Solar Panels to the SP25. This is the only source of power to the SP25.
- B. Power output from the SP25 used to charge batteries.

SP25 Operation

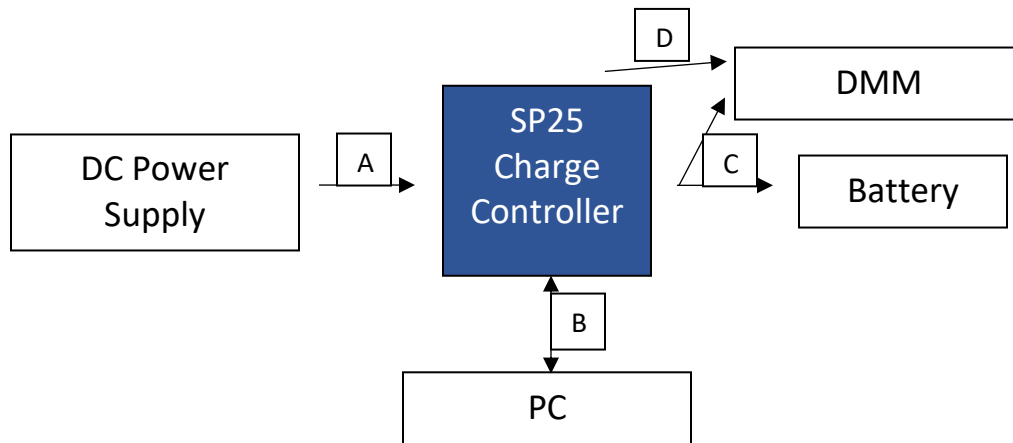
The SP25 is powered on when anything above 12V is applied on input A. After receiving power, it takes the SP25 3+ seconds to become operational. Once operational the SP25 will begin to deliver power on output B.

The SP25 also includes an interface for configuring the chip for customers. A PC can be hooked up to it using USB, and through software shipped by the supplier: set and read certain configurations. The specific configurations of the controller are as follows

1. Read/Write to set the output for 12V, or 24V mode. This changes line B on Figure 1.
2. Read/Write the fan to turn on at a certain temperature.
3. Read the status of the SP25.

SP25 Automated Testing Hardware Overview

The SP25 within the test configuration looks like this:



- A. Input lines from the DCPowerSupply to the SP25: used to imitate solar panels.
- B. Input/output lines from the PC to the SP25: used to communicate with the device.
- C. Output lines to a battery, also connected to the DMM. These lines are used for voltage/current reading.
- D. Output lines from the SP25 to the DMM. These lines are used for temperature, and fan information.

The SP25 is powered on when anything above 12V is applied on input A. After receiving power, it takes the SP25 at least 3 seconds for the output to relate to the input. Once operational the SP25 will begin to deliver output on the lines C and D. While the device is connected to the PC, even if power is off, communication can happen.

DMM:

There are 9 active lines on the DMM. All of them are DC lines. These lines are only “active” for 10 seconds after you receive a “Ready” from the SP25 Status. After the 10 seconds, the lines will only report 0’s for voltages and currents.

The SP25 provides 8 status LED’s. The DC lines (0-7) of the DMM are connected to these 8 LED’s. Lines 0-6 are connected to the LED’s that correspond to the current temperature status of the SP25 charge controller. Line 7 is connected to the LED that indicates the current fan status (ON/OFF) of the SP25 charge controller.

Corresponding Temperature Lines (0-6) are represented in the table below

Line 0	Line 1	Line 2	Line 3	Line 4	Line 5	Line 6
0 – 9.99 C	10 – 19.99 C	20 – 29.99 C	30 – 39.99 C	40 – 49.99 C	50 – 59.99 C	60 – 69.99 C

The current temperature of the SP25 is represented by the respective LED being in the ON state (Digital Voltage $\geq 1V$). Only one line shall be in the ON state. The remaining lines shall be in the OFF state (0V).

When the Fan is in the ON state, the DC Voltage of Line 7 shall be 5V ($\pm 0.1V$), and the DC Current shall be 0.4A ($\pm 0.05A$). When the Fan is in the OFF state, the DC Voltage shall be 0V and the DC Current shall be 0A.

Line 8 of the DMM is connected to the output from the SP25 to the battery. When the SP25 returns a status of "Ready" the output to the battery is enabled. The DC Voltage and Current shall be determined by the SP25 Mode Setting and the DC Power Supply setting; the Test Specifications below shall provide the corresponding settings and the expected outputs.

VM1000 Test Specification

Using the APIs (found in Appendix A, B, and C) for the SP25 and instruments: Create an automated test to verify that the SP25 Charge Controller is fully functional and is sending/receiving signals properly. This will consist of ONE sequence file that contains all your code. Follow the bullets below to test at certain configurations and TestStand settings.

- TestStand Settings
 - Allow Tracing into Setup/Cleanup.
 - Allow Tracing While Terminating.
 - Tracing speed set to slowest possible.
 - Run-Time Error should show a Dialog Box.
 - Allow Prompt to Find Files.
 - Add a search directory for the location of the code modules
"C:\Users\Public\Documents\CTD_Test". This should search the subdirectories.
- Instrument Setup
 - Initialize the DCPowerSupply and set the current limit to 15A.
 - Initialize the DMM
 - Initialize the SP25 connection.
 - Ensure that the SP25 reports "OFF" as the status.
- Test 1
 - Turn off the output of the DCPowerSupply
 - Set the Fan to operate at 50-59.99C.
 - Read the fan to check if it got set correctly.
 - Set the mode to 12V.
 - Read the mode to check if it got set to 12V correctly.
 - Set the Voltage on the DCPowerSupply to 15V.
 - Turn on the output of the DCPowerSupply
 - Wait until the Status says "Ready"
 - Check that the temperature is 30-39.99C.
 - Check that the fan is off, check voltage and current.
 - Check that the output to the battery is 14V (± 0.3), 2A (± 0.05).
- Test 2
 - Turn off the output of the DCPowerSupply.
 - Set the Fan to operates at 30-39.99C.
 - Read the Fan to check if it got set correctly.
 - Set the mode to 12V.
 - Read the mode to check if it got set to 12V correctly.
 - Set the Voltage on the DCPowerSupply to 20V.
 - Turn on the output of the DCPowerSupply

- Wait until the Status says “Ready”
- Check that the temperature is 30-39.99C.
- Check that the fan is on, check voltage and current.
- Check that the output to the battery is 14V (+- 0.3), 4A (+- 0.05).
- Test 3
 - Turn off the output of the DCPowerSupply.
 - Set the Fan to operate at 40- 49.99C.
 - Read the Fan to check if it got set correctly.
 - Set the mode to 24V.
 - Read the mode to check if it got set to 24V correctly.
 - Set the Voltage on the DCPowerSupply to 30V.
 - Turn on the output of the DCPowerSupply
 - Wait until the Status says “Ready”
 - Check that the temperature is 40-49.99C.
 - Check that the fan is on, check voltage and current.
 - Check that the output to the battery is 27V (+- 0.3), 3A (+- 0.05).
- If the test fixture cannot be initialized
 - Suppress all errors (but mark the test as an error)
 - Record that it could not initialize the test fixture using an Additional Results step.

VM1000 Automated Testing Software Overview

All APIs are supplied to you and can be found in C:\Users\Public\Documents. The following Appendix additions explain each function that can be used with each instrument. At the end of each table of explanations is a table of error codes and what they mean.

Appendix A: SP25 API

This API oversees the communication between the PC and the SP25 Circuit Card.

To initialize the connection, you must pass in the correct information on the Initialize.vi Name parameter. **Name** must be “SP25”. After this you just need to hold onto the reference number.

Function Name	Parameter Name	Direction	Type	Purpose	Errors
Initialize	Name	In	String	The name of the instrument. Internally this creates a reference to the instrument.	5001
	Reference Out	Out	U32	Numerical reference to the instrument.	
	Success	Out	Boolean	Indicates whether initialization was successful or not.	
Write Mode	Reference In	In	U32	Numeric reference to the instrument.	5001, 5002

	24V Mode	In	Boolean	Indicates whether to turn on 24V Mode. False = Turn Off, True = Turn On.	
	Reference Out	Out	U32	Numeric reference to the instrument.	
Read Mode	Reference In	In	U32	Numeric reference to the instrument.	5001
	Reference Out	Out	U32	Numeric reference to the instrument.	
	24V Mode	Out	Boolean	Indicates whether the 24V Mode is on. True = On, False = Off.	
Read Status	Reference In	In	U32	Numeric reference to the instrument.	5001
	Reference Out	Out	U32	Numeric reference to the instrument.	
	Status	Out	String	Indicates the status of the SP25. Possible answers are “Ready”, “Preparing”, “Off”	
Write Fan	Reference In	In	U32	Numeric reference to the instrument.	5001, 5002
	Fan	In	Enum	Indicates what temperature to turn the fan on at: 0-9.99C, 10-19.99, 20-29.99, 30-39.99, 40-49.99, 50-59.99, 60-69.99.	
	Reference Out	Out	U32	Numeric reference to the instrument.	
Read Fan	Reference In	In	U32	Numeric reference to the instrument.	5001
	Reference Out	Out	U32	Numeric reference to the instrument.	
	Fan	Out	Enum	Indicates what temperature the fan will turn on at:	

				0-9.99C, 10-19.99, 20-29.99, 30-39.99, 40-49.99, 50-59.99, 60-69.99.	
Close	Reference In	In	U32	Numeric reference to the instrument.	5001
	Success	Out	Boolean	Indicates whether close was successful or not.	

Error Codes and Explanations

5000	Boot error, instrument is not initialized.
5001	For Initialize, if Name passed in does not correspond to the instrument.
5002	The reference provided is invalid.
5003	Trying to change configuration while input has power.

Appendix B: DMM API

To initialize the DMM you must pass in the correct information on the Initialize.vi Visa resource name in parameter. **DeviceName** must be "COM1".

Function Name	Parameter Name	Direction	Type	Purpose	Errors
Initialize	DeviceName	In	String	Used to tell the function what alias the power supply uses to establish a connection.	5010, 5013
	Visa resource name out	Out	VISA	The VISA resource after opening the connection to the instrument.	
	Success	Out	Boolean	True if successfully connected	
Get DC Voltage	Visa resource name in	In	VISA	The active VISA resource connection.	5010, 5011
	Channels	In	Array of U8	Channels to read.	
	Visa resource name out	Out	VISA	The active VISA resource connection.	
	Voltages (Volts)	Out	Array of Double	Voltages on the channels passed in	

Get DC Current	Visa resource name in	In	VISA	The active VISA resource connection.	5010, 5011
	Channels	In	Array of U8	Channels to read.	
	Visa resource name out	Out	VISA	The active VISA resource connection.	
	Currents (Amps)	Out	Array of Double	Currents on the channels passed in.	
Get AC Voltage	Visa resource name in	In	VISA	The active VISA resource connection.	5010, 5012
	Channel	In	U8	Channel to read the voltage on	
	Visa resource name out	Out	VISA	The active VISA resource connection.	
	Voltage (Volts)	Out	Double	Voltage of the channel	
Get AC Current	Visa resource name in	In	VISA	The active VISA resource connection.	5010, 5012
	Channel	In	U8	Channel to read the current on	
	Visa resource name out	Out	VISA	The active VISA resource connection.	
	Current (Amps)	Out	Double	Current for the channel	
Close	Visa resource name in	In	VISA	The active VISA resource connection.	5010
	Success	Out	Boolean	Returns true if the DMM was shut down correctly.	

Error Codes and Explanations

5010	Instrument is not initialized.
5011	The channel given is not a DC channel
5012	The channel given is not an AC channel
5013	When calling initialize, an instrument with that name does not exist

Appendix C: Power Supply API

To initialize the Power Supply you must pass in the correct information on the Initialize.vi Visa resource name in parameter. **DeviceName** must be "COM2".

Function Name	Parameter Name	Direction	Type	Purpose	Errors
Initialize	DeviceName	In	String	Used to tell the function what alias the power supply uses to establish a connection.	5020, 5021
	Visa resource name out	Out	VISA	The VISA resource after opening the connection to the instrument.	
	Success	Out	Boolean	True if successfully connected	
Set Current Limit	Visa resource name in	In	VISA	The active VISA resource connection.	5020, 5023
	Current Limit (Amps)	In	Number	The limit at which the power supply will stop supplying current to the UUT.	
	Visa resource name out	Out	VISA	The active VISA resource connection.	
Set Voltage	Visa resource name in	In	VISA	The active VISA resource connection.	5020, 5022
	Voltage (Volts)	In	Double	The voltage for the DC Power Supply	
	Visa resource name out	Out	VISA	The active VISA resource connection.	
Set Output State	Visa resource name in	In	VISA	The active VISA resource connection.	5020
	State	In	Boolean	If True the DC Power Supply will turn the voltage on to the UUT. If false no current or voltage will be emitted from the DC Power Supply	
	Visa resource name out	Out	VISA	The active VISA resource connection.	
Get Voltage And Current	Visa resource name in	In	VISA	The active VISA resource connection.	5020
	Visa resource name out	Out	VISA	The active VISA resource connection.	

	Voltage (Volts)	Out	Double	Returns the voltage being output.	
	Current (Amps)	Out	Double	Returns the current being drawn	
Get Current Limit	Visa resource name in	In	VISA	The active VISA resource connection.	5020
	Visa resource name out	Out	VISA	The active VISA resource connection.	
	Current Limit (Amps)	Out	Double	Returns what the current limit is set to.	
Close	Visa resource name in	In	VISA	The active VISA resource connection.	5020
	Success	Out	Boolean	Returns True if the DC Power Supply was shut down correctly.	

Error Codes and Explanations

5020	Instrument is not initialized.
5021	When calling initialize, an instrument with that name does not exist
5022	The user tries to set the voltage to some number out of bounds
5023	The user tries to set the current to some number out of bounds